

Yep. Let's make this **much smaller and sequential**. You don't need to understand the whole phase vocoder at once.

Phase Vocoder Learning Roadmap

Go through these questions **in order**. If you can answer each one, you'll naturally arrive at the implementation.

● Level 1 — FFT representation

You already mostly have this.

1. What does an FFT bin represent?
 2. Why is bin k associated with frequency kF_s/N ?
 3. What does the **magnitude** of an FFT bin tell us?
 4. What does the **phase** of an FFT bin tell us?
 5. Why does a real-valued signal have positive **and negative frequencies**?
 6. What happens in the FFT when a sinusoid's frequency lies **between two bins**?
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● Level 2 — Phase

This is the important prerequisite.

7. For

$$x(t) = A \cos(2\pi ft + \phi)$$

what does ϕ physically mean?

8. If a sinusoid is oscillating at frequency f , how does its phase change with time?
9. If phase changes by 2π , what happened physically?
10. If phase changes by 4π in one second, what frequency does that correspond to?
11. **Can you determine a sinusoid's frequency from how quickly its phase changes?**

That last question is the bridge into the phase vocoder.

● Level 3 — STFT over time

Now take what you know about FFT and apply it to multiple windows.

12. Why does STFT produce a sequence of FFTs instead of one FFT?
13. What does:

$$X[k, n]$$

mean?

14. What does k represent?
15. What does n represent?
16. If the same 440 Hz sinusoid appears in consecutive windows, what happens to its **magnitude**?
17. What happens to its **phase**?
18. **How much should the phase advance between two consecutive windows?**

This is probably the most important question in the whole roadmap.

● Level 4 — Frequency from phase

Now we answer:

"How can I figure out the actual frequency of a component?"

19. What phase advance should I expect for FFT bin k ?
20. What happens if the actual sinusoid's frequency is slightly different from the bin frequency?
21. How can the difference between **expected phase advance** and **measured phase advance** reveal the actual frequency?
22. Why do we need to deal with phase wrapping?

For example:

$$6.1 \rightarrow 0.1$$

Does that mean the phase went backwards by 6 radians?

● Level 5 — Time stretching

Only **after** understanding the above.

23. What happens if I make the distance between synthesis frames larger than the distance between analysis frames?
24. Why does that make the audio longer?
25. If I simply spread the frames apart, why might the pitch change?
26. How can I modify the phase progression so that the original frequencies remain unchanged?
27. What does **phase accumulation** mean?

At this point you basically understand the core phase-vocoder idea.

🔴 Level 6 — Pitch shifting

Now come back to your original question.

28. What is the difference between **time stretching** and **pitch shifting**?

29. If I want:

$$440 \rightarrow 880 \text{ Hz}$$

what should happen to the spectral components?

30. Why can't I simply change the frequency labels of FFT bins?

31. Why do I need to **move/interpolate spectral energy**?

32. After moving a component to a new frequency, how should its phase evolve?

33. Why does independently processing every window produce discontinuities?

34. How does phase tracking prevent those discontinuities?

🧠 The entire thing in one chain

Eventually you want to be able to reason through this:

```
FFT
↓
"I have magnitude + phase"
↓
Phase changes over time
↓
"Phase change tells me frequency"
↓
Track frequency across STFT frames
↓
Manipulate timing/frequency
↓
Adjust phase accordingly
↓
IFFT
↓
Overlap-add
```

Don't study all 34 at once.

I'd actually recommend doing:

1–6 → stop

Then:

7–11 → stop

Then:

12–18 → stop

And so on.

And based on where you are right now, **the next thing I'd tackle is #7:**

| For $x(t) = A \cos(2\pi ft + \phi)$, what does ϕ physically mean?

Once that clicks, the rest of the phase stuff becomes way less intimidating.